

Cogni Map: Real-Time Detection of Cognitive Actions in Language Models Through Linear Probing

Ivan Chulo, Ananya Joshi

Johns Hopkins University
Baltimore, MD 21218 USA

Abstract

Theory of Mind (ToM)—the ability to reason about others’ mental states—is crucial for AI alignment and human-AI interaction, yet its mechanistic implementation in language models remains opaque. We introduce Cogni Map, a mechanistic interpretability framework that decomposes ToM into constituent cognitive processes using linear probes trained on 45 cognitive actions. By combining probe-based detection with Contrastive Activation Addition (CAA) steering vectors trained on perceptual contrasts from the BigToM benchmark, we achieve both ToM performance improvements (45% to 56% accuracy) and mechanistic insights: steering systematically increases perceptual attention (*emotion_perception*, *noticing*) and hypothesis formation while reducing explicit interrogation. This demonstrates how cognitive action decomposition can bridge representation engineering and cognitive science, offering interpretable analysis of high-level AI capabilities.

Introduction

Theory of Mind (ToM)—the capacity to attribute mental states to oneself and others—represents a critical capability for AI systems engaged in social reasoning, alignment, and human collaboration (Bereska and Gavves 2024). While recent work has demonstrated that large language models (LLMs) can perform ToM tasks (Chen et al. 2024), the *mechanistic basis* of this ability remains unclear: What cognitive processes constitute ToM reasoning? How do internal representations shift when perspective-taking is engaged?

Traditional evaluations treat ToM as a monolithic capability measured through binary accuracy on belief attribution tasks. However, ToM in cognitive science comprises multiple interacting processes: perceptual encoding, hypothesis generation, emotional inference, and counterfactual reasoning. Understanding which components are active—and how steering vectors modulate them—would enable principled improvement of AI social cognition and earlier detection of reasoning failures.

We introduce Cogni Map, a mechanistic interpretability framework combining two techniques: (1) linear probes trained on a 45-action cognitive taxonomy to detect active

reasoning processes, and (2) Contrastive Activation Addition (CAA) steering vectors to enhance ToM while measuring resulting cognitive shifts. This approach moves beyond “does the model have capability X?” to “how is capability X mechanistically implemented?”

Contributions: (1) A cognitive action taxonomy with 45 categories and 31,500 training examples; (2) Binary probes achieving 0.78 AUC-ROC with layer specialization peaking at layer 9; (3) CAA steering vectors trained on perceptual contrasts from BigToM, improving accuracy 11 percentage points; (4) Mechanistic decomposition revealing ToM operates through perceptual attention and hypothesis formation rather than explicit interrogation.

Methodology

Cognitive Action Taxonomy. A taxonomy of 45 cognitive actions was defined across four categories: *Metacognitive* (e.g., *reconsidering*), *Analytical* (e.g., *analyzing*), *Creative* (e.g., *reframing*), and *Emotional* (e.g., *emotional_reappraisal*).

Activation Capture and Probing. Following established probing methodologies (Alain and Bengio 2016; Chen et al. 2024), activations were extracted from 30 of the 34 layers of Gemma-3-4B using nnsight. During both activation capture and inference, inputs were augmented with the suffix “The cognitive action being demonstrated here is” to ensure consistent extraction from the final token representation. A **one-vs-rest** strategy was used to train 45 independent binary linear probes, which allows for per-action interpretability and the flexibility to mix optimal layers for each action during inference. Training was performed with an AdamW optimizer, cosine annealing scheduler, and early stopping, using an AUC-ROC metric to handle the severe class imbalance.

CAA Steering Vector Training. To isolate ToM-specific cognitive processes, we trained Contrastive Activation Addition (CAA) steering vectors (Rimsky et al. 2023) using the representation engineering framework (Zou et al. 2023). We constructed 752 contrastive triplets from BigToM’s *forward_belief* scenarios (both true and false belief conditions). Each triplet contains: (1) story context with observational information (protagonist sees or does not see an event), (2) belief attribution question, (3) positive completion: correct belief attribution, (4) negative completion: incorrect belief

attribution. This captures the representational difference between accurate and inaccurate ToM reasoning rather than surface-level observational statements. Vectors were trained across layers 14-30 using PCA-centered activation differences, exported in GGUF format, and applied with coefficient=1.5.

ToM Decomposition Methodology. We evaluated ToM reasoning on 100 samples from the BigToM procedural benchmark, comparing baseline (no steering) vs. steered conditions. For each sample containing a story, question, true answer, and wrong answer, we captured cognitive action activations at three timepoints: (1) after story+question (before answer), (2) after story+question+true answer, and (3) after story+question+wrong answer. Answer selection used probability ranking over multiple-choice letters (a/b), following BigToM methodology. For each cognitive action, we computed layer count (number of layers 10-20 where probe confidence exceeded threshold 0.001) and analyzed differences between baseline and steered conditions to identify which cognitive processes are modulated by ToM enhancement.

Results

Probe Performance. Binary probes achieved 0.78 average AUC-ROC and 0.68 F1 across all 45 actions, confirming linear separability of cognitive representations (Alain and Bengio 2016). Performance ranged from *suspending_judgment* (0.988 AUC) and *counterfactual_reasoning* (0.984 AUC) to *emotion_responding* (0.778 AUC). Layer 9 showed peak performance (0.948 AUC), with strong mid-layer performance (layers 5-24) and degraded early/late layers, suggesting early layers encode low-level features while mid-layers capture semantic abstractions. This informed our layer 10-20 analysis window.

ToM Steering Efficacy. CAA steering (coefficient=1.5) on 100 BigToM samples improved accuracy from 45% to 56%. Correct answer probability increased from 0.446 to 0.560, while incorrect decreased from 0.554 to 0.440, confirming perceptual contrast training enhances ToM without explicit prompting.

Cognitive Decomposition of ToM. ToM steering systematically modulated cognitive action patterns. At question timepoint: *emotion_perception* (+2.35 layer count), *noticing* (+1.59), *hypothesis_generation* (+1.57), *emotion_valuing* (+0.94), *suspending_judgment* (+0.82), *questioning* (-1.01). After true answer: *emotion_perception* (+1.48), *noticing* (+1.24), *hypothesis_generation* (+1.18), *suspending_judgment* (+1.09), *convergent_thinking* (-1.48), *questioning* (-1.27), *understanding* (-1.02). Patterns indicate heightened perceptual attention and hypothesis formation with reduced explicit interrogation.

[PLACEHOLDER: Full-Scale Results.] The complete evaluation across all 12 BigToM conditions (forward/backward belief/action × true/false belief) and systematic analysis of layer-wise activation patterns, timepoint dependencies, and cognitive action correlations are currently in progress. Preliminary results suggest condition-specific cognitive profiles, with false belief tasks requiring enhanced

counterfactual_reasoning and backward inference tasks engaging *metacognitive_monitoring*.

Discussion

Mechanistic Insights. Our decomposition reveals ToM operates through perceptual grounding rather than deliberate reasoning. Increases in *emotion_perception* and *noticing* coupled with decreases in *questioning* and *convergent_thinking* suggest a shift from effortful "cold" cognition to automatic "hot" social cognition—challenging assumptions that LLM ToM relies primarily on chain-of-thought reasoning. This mirrors cognitive science distinctions between explicit and implicit perspective-taking.

Methodological Contribution. Using perceptual contrasts ("sees event" vs. "does not see event") to train CAA vectors isolates ToM-specific processing without instruction contamination, enabling precise attribution of cognitive shifts to perspective-taking mechanisms rather than general task improvement (Rimsky et al. 2023). This approach generalizes to other capabilities requiring mechanistic decomposition.

Limitations. Synthetic probe training data, single model evaluation (Gemma-3-4B), preliminary single-condition results (100 *forward_belief_true_belief* samples), independence assumptions between cognitive actions, and layer count metrics that may not capture activation magnitude.

Future Work. Full 12-condition BigToM evaluation, layer-wise magnitude analysis, temporal dynamics across measurement timepoints, extension to commonsense reasoning/moral judgment/deception detection, human annotation validation, and scaling to larger models.

Broader Impact. Process-based supervision via cognitive monitoring enables early detection of reasoning failures for AI safety. Understanding constituent processes of capabilities like deception or persuasion allows targeted steering interventions for alignment. Applications in education and mental health require responsible deployment guidelines to prevent manipulation via emotional optimization.

Conclusion

Cogni Map demonstrates that complex AI capabilities can be mechanistically decomposed into interpretable cognitive components. Our framework combines linear probing for process detection with CAA steering for targeted intervention, revealing that ToM enhancement operates through perceptual grounding and implicit inference rather than deliberate reasoning. This bridges mechanistic interpretability and cognitive science, offering principled methodology for understanding and steering AI reasoning processes. The toolkit, probes, and datasets enable extension to other high-level capabilities including commonsense reasoning, moral judgment, and deception detection.

References

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Appendix

CAA Training Data Example

The CAA steering vectors were trained on contrastive pairs distinguishing correct from incorrect belief attribution. Each training triplet consists of a story with observational information, a question about the protagonist’s belief, and two completions representing accurate vs. inaccurate Theory of Mind reasoning.

False Belief Example:

Story: “Noor is working as a barista at a busy coffee shop. Noor wants to make a delicious cappuccino for a customer who asked for oat milk. Noor grabs a milk pitcher and fills it with oat milk. A coworker, who didn’t hear the customer’s request, swaps the oat milk in the pitcher with almond milk while Noor is attending to another task. Noor does not see her coworker swapping the milk.”

Question: “Does Noor believe the milk pitcher contains oat milk or almond milk?”

Positive (correct ToM): “Noor believes the milk pitcher contains oat milk.”

Negative (incorrect ToM): “Noor believes the milk pitcher contains almond milk.”

True Belief Example:

Story: “Noor is working as a barista at a busy coffee shop. Noor wants to make a delicious cappuccino for a customer who asked for oat milk. Noor grabs a milk pitcher and fills it with oat milk. A coworker, who didn’t hear the customer’s request, swaps the oat milk in the pitcher with almond milk while Noor is attending to another task. Noor sees her coworker swapping the milk.”

Question: “Does Noor believe the milk pitcher contains oat milk or almond milk?”

Positive (correct ToM): “Noor believes the milk pitcher contains almond milk.”

Negative (incorrect ToM): “Noor believes the milk pitcher contains oat milk.”

The key distinction is that the positive completion always reflects accurate perspective-taking (understanding what the protagonist actually believes based on their observational access), while the negative completion represents a ToM failure (confusing the protagonist’s belief with objective reality or making incorrect inferences).

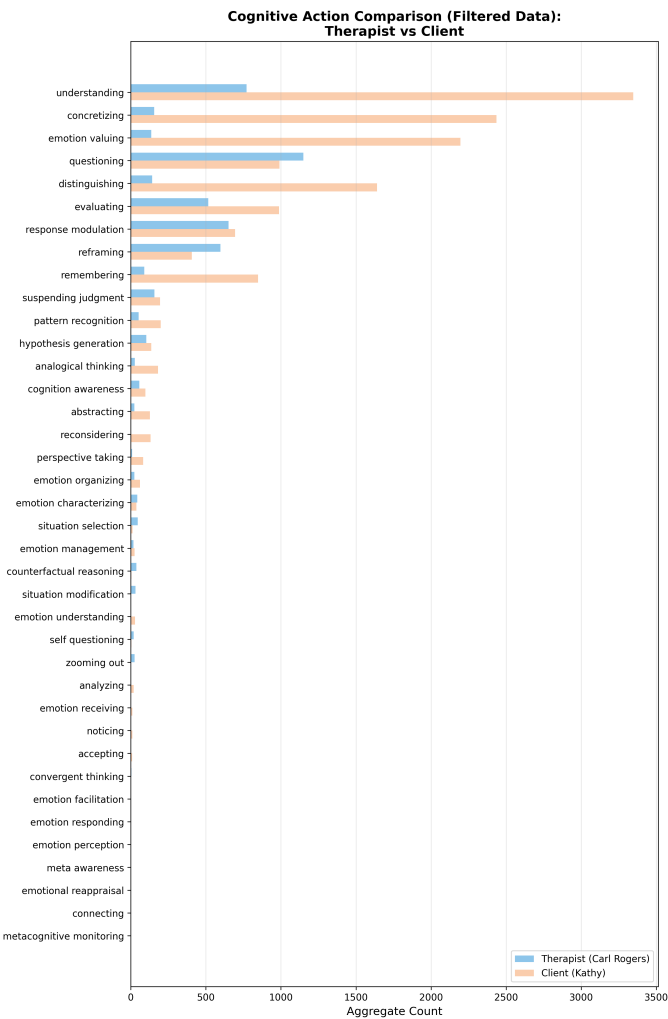


Figure 1: Qualitative validation of cognitive probe framework through comparison of therapist (Carl Rogers) and client (Kathy). Left panel shows aggregate counts, highlighting role-specific patterns (therapist: *situational awareness, self-questioning*) consistent with person-centered therapy principles. Right panel shows log2 ratio bias, highlighting interpretability beyond controlled ToM benchmarks.

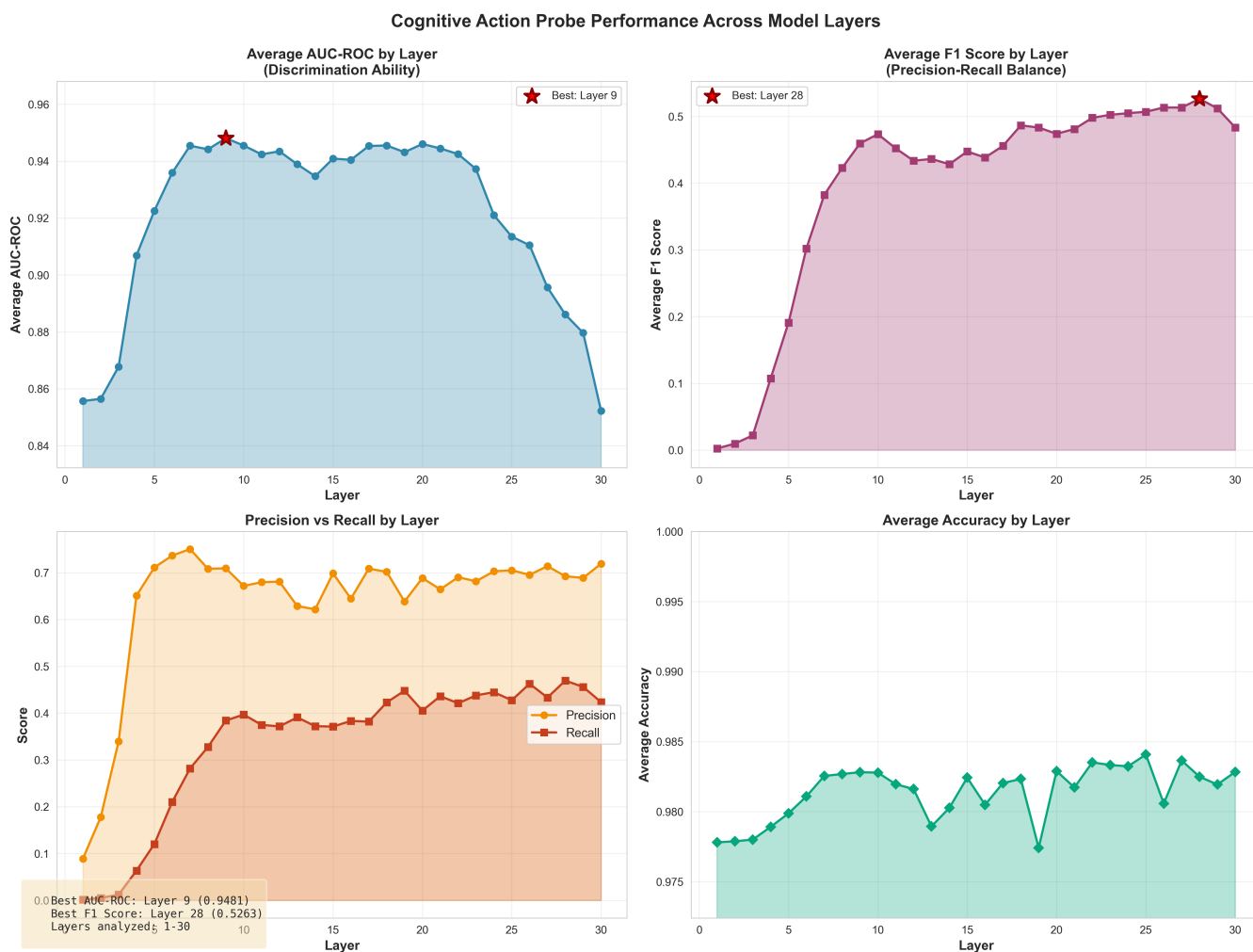


Figure 2: Cognitive action probe performance across all 30 layers of Gemma-3-4B. The visualization shows average AUC-ROC scores for each layer, with Layer 9 achieving peak performance (0.948 AUC-ROC). Strong performance is maintained across mid-layers (5-24), while early and late layers show degraded performance. This pattern suggests that early layers focus on surface-level features, mid-layers capture high-level cognitive abstractions, and late layers optimize for next-token prediction, potentially overwriting intermediate representations.

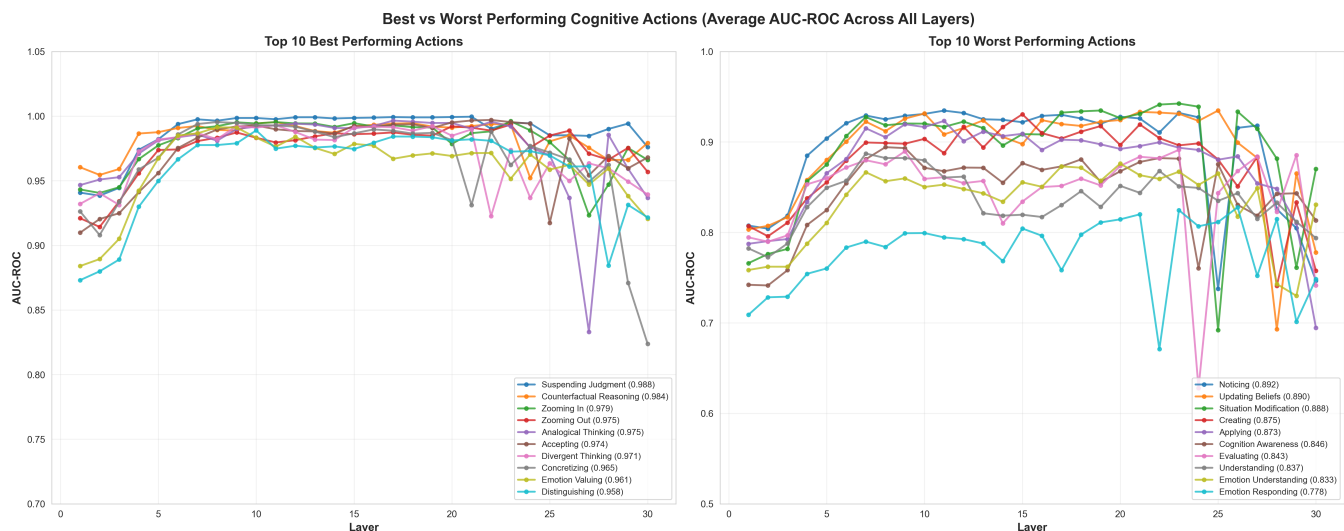


Figure 3: Comparison of top 10 and bottom 10 performing cognitive actions ranked by average AUC-ROC across all layers. Best performers like *suspending_judgment* (0.988) and *counterfactual_reasoning* (0.984) show consistently high performance and distinct activation patterns across most layers. Worst performers like *emotion_responding* (0.778) and *understanding* (0.837) exhibit more variability and lower overall discrimination ability, suggesting these concepts may be more distributed or context-dependent in the model’s representation space.

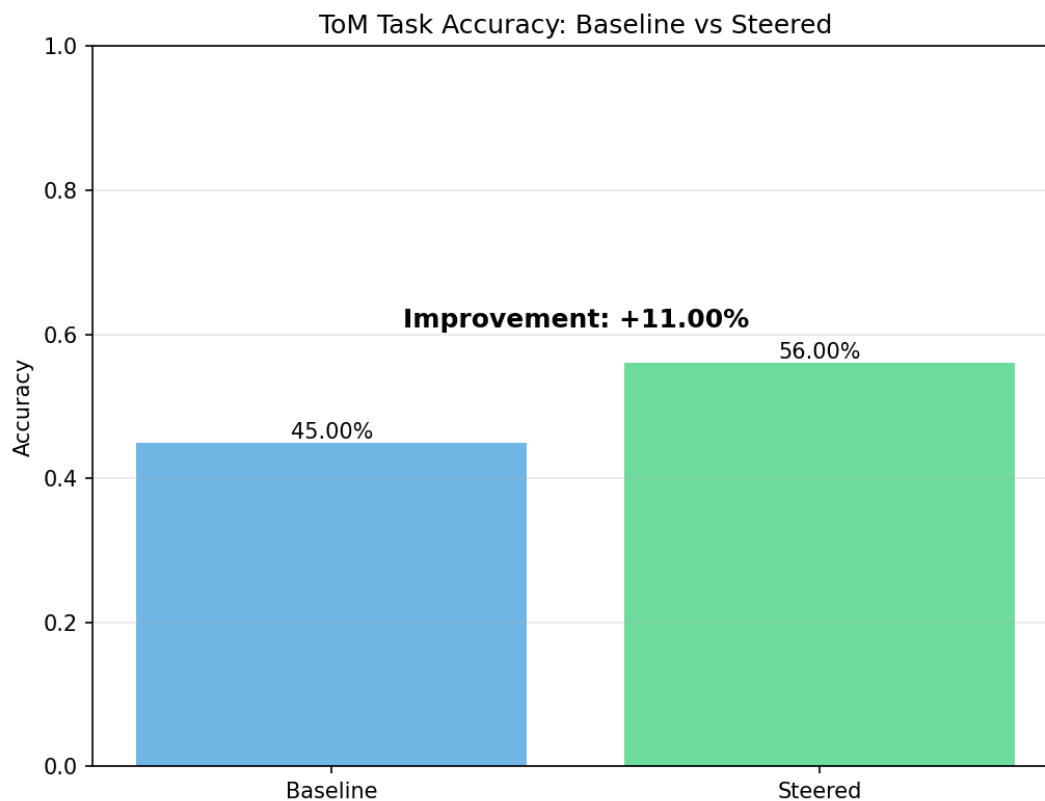


Figure 4: [PRELIMINARY] ToM steering efficacy on 100 BigToM samples from *forward_belief_true_belief* condition. Baseline accuracy (no steering) was 45%, while CAA-steered inference (coefficient=1.5) achieved 56% accuracy, representing an 11 percentage point improvement. Error bars indicate 95% confidence intervals. Full evaluation across all 12 BigToM conditions pending.

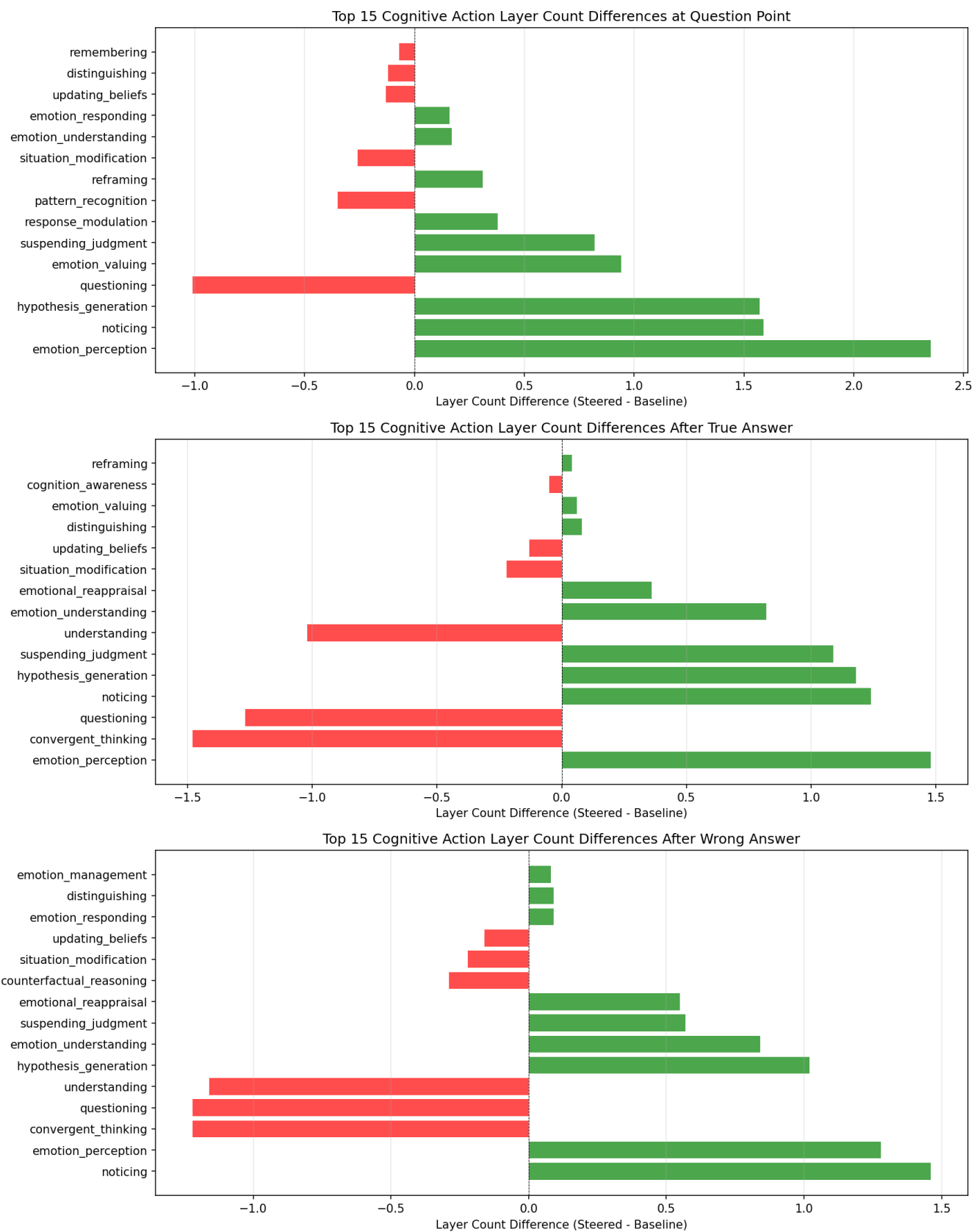


Figure 5: [PRELIMINARY] Cognitive action activation differences between steered and baseline conditions at question time-point. Positive values indicate increased layer count under ToM steering. Top increases: *emotion_perception* (+2.35), *noticing* (+1.59), *hypothesis_generation* (+1.57), *emotion_valuing* (+0.94), *suspending_judgment* (+0.82). Notable decrease: *questioning* (-1.01). Error bars show standard error across 100 samples.

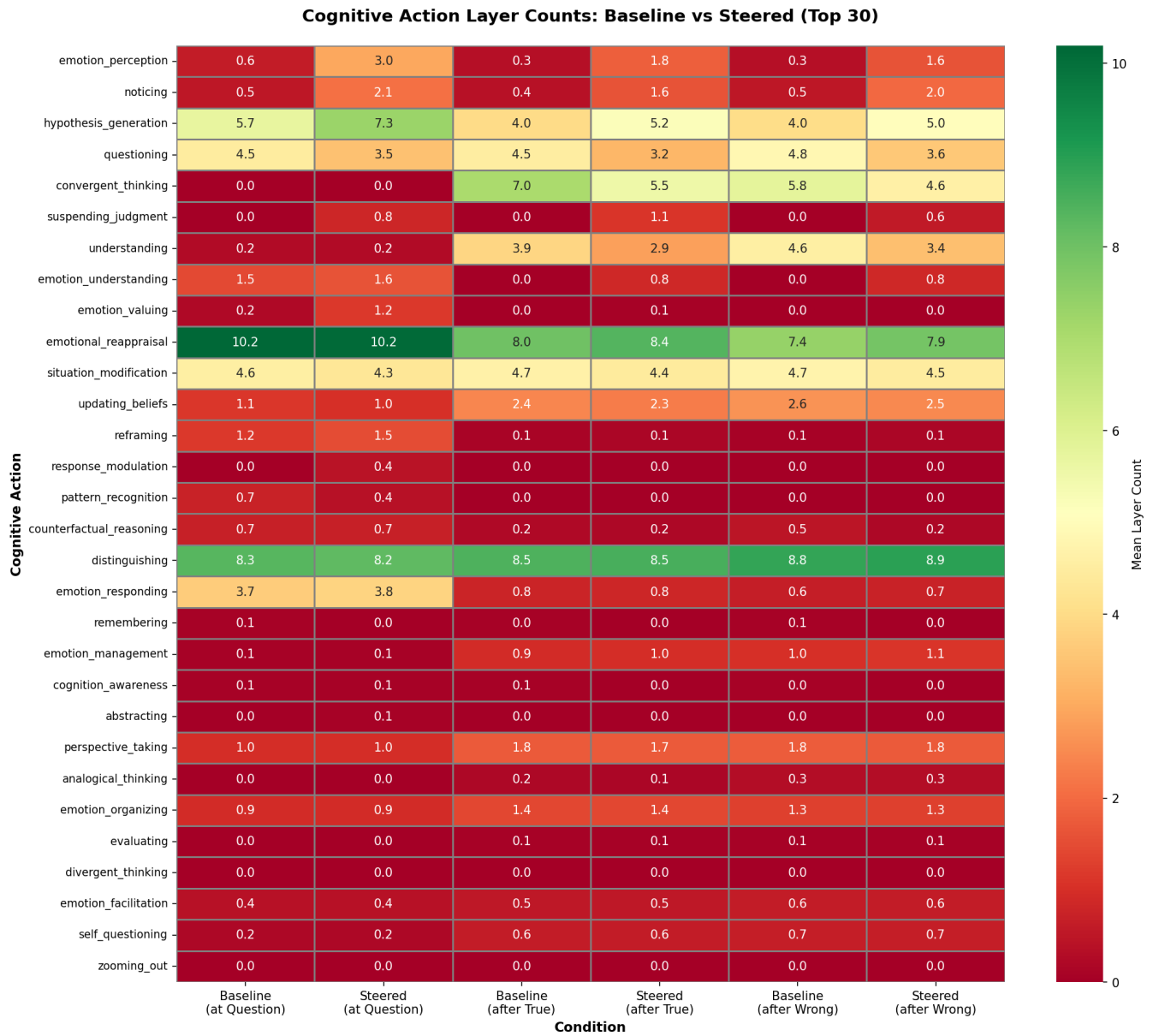


Figure 6: [PRELIMINARY] Heatmap of cognitive action activation differences across three timepoints: at question (before answer), after true answer, and after wrong answer. Rows represent cognitive actions, columns represent measurement timepoints. Color intensity indicates magnitude of change (red = increase, blue = decrease). Consistent patterns across timepoints (e.g., *emotion_perception*, *noticing*) suggest robust ToM-related cognitive processes, while timepoint-specific patterns reveal dynamic reasoning trajectories.